

Marketing Mix Modeling (MMM)

DT MART

Measuring the Impact of Marketing Channels on Sales Revenue

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Date: May 2026

Dataset: DT MART Market Mix Modeling (Kaggle)

Tools: Python, Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, Seaborn

1. Executive Summary

This report presents a Marketing Mix Model (MMM) built for DT MART, a retail brand, using 12 months of marketing and sales data from July 2015 to June 2016. The model was built in Python using OLS regression to quantify the revenue contribution of each marketing channel, identify points of diminishing returns, and recommend an optimized budget allocation.

The model achieved an R-squared of 0.977 and an Adjusted R-squared of 0.950, confirming strong predictive accuracy. Key findings show that Affiliates is the most cost-efficient channel, Sponsorship is significantly over-funded relative to its revenue contribution, and Online Marketing delivers reliable returns at current spend levels.

2. Objective

The primary objectives of this project are:

- Quantify the revenue contribution of each marketing channel
- Identify channels that are approaching diminishing returns
- Build a regression model to predict total sales revenue from marketing inputs
- Recommend an optimized budget allocation to improve return on investment

3. Dataset Overview

The dataset was sourced from Kaggle (DT MART Market Mix Modeling) and consists of seven CSV files capturing different aspects of marketing and sales activity.

File	Rows	Description
ProductList	75	Product names with frequency and share percentage
SpecialSale	44	Dates and names of special sale events
MediaInvestment	12	Monthly marketing spend by channel (2015)
FirstFile	1,578,079	Transaction-level sales with product and discount details
SecondFile	12	Monthly aggregated revenue, units, discounts, media spend, and NPS
Sales	1,048,575	Raw transaction data (tab-separated)
MonthlyNPS	12	Monthly Net Promoter Scores

The primary modeling file was SecondFile, which provides clean monthly aggregated data combining revenue across five product categories, all marketing channel investments, and NPS scores. SpecialSale was used to engineer a promotional days feature.

4. Methodology

4.1 Data Cleaning

The dataset required minimal cleaning. Two columns had missing values:

- Radio: 9 missing values out of 12 months
- Other: 9 missing values out of 12 months

Both columns were filled with zero since no spend was recorded for those periods. The Date column was parsed to datetime format and the dataset was sorted chronologically.

4.2 Feature Engineering

The following features were engineered before model building:

- Log transformation applied to the top four marketing channels (Online Marketing, Affiliates, Sponsorship, TV) to capture diminishing returns effects
- A promotional days flag was created by merging the SpecialSale dates with the monthly dataset to count the number of promotional days in each month
- NPS score was retained as a direct feature representing customer satisfaction

4.3 Model Building

An OLS (Ordinary Least Squares) regression model was built using Statsmodels. The final model used six features: log-transformed values of Online Marketing, Affiliates, Sponsorship, and TV spend, along with NPS score and promotional days. Feature selection was performed by removing channels with high p-values and multicollinearity to arrive at a reduced model.

4.4 Contribution Analysis

Channel revenue contributions were calculated by multiplying each model coefficient against its corresponding log-transformed feature values across all observations and summing the result. Contributions were expressed as a percentage of total model-explained revenue.

4.5 Diminishing Returns Modeling

A logarithmic curve was fitted to each top channel by regressing actual spend against actual revenue. These saturation curves were used to visually identify where incremental spend yields rapidly declining revenue gains.

4.6 Budget Optimization

An efficiency score was calculated for each channel as the ratio of revenue contribution percentage to current spend percentage. Budget was then reallocated proportionally based on these efficiency scores, keeping total budget constant.

5. Exploratory Data Analysis

5.1 Revenue Trend

Monthly total revenue ranged from near zero in August 2015 to a peak of approximately 50 Cr in October 2015. The August data point is a clear anomaly, likely due to incomplete transaction recording. From January 2016 onward, revenue stabilized in the 30 to 40 Cr range with no strong growth trend over the period.

5.2 Marketing Investment by Channel

Sponsorship was the dominant channel in every month, accounting for the largest share of total investment. October 2015 recorded the highest total marketing spend at approximately 1700M, which aligned with the peak revenue of the same month. August 2015 had the lowest investment and correspondingly the lowest revenue, reinforcing the relationship between spend and output.

5.3 Channel Correlation with Revenue

Channel	Pearson Correlation	Interpretation
Online Marketing	0.89	Very strong positive relationship
Affiliates	0.87	Very strong positive relationship
Sponsorship	0.73	Strong positive relationship
TV	0.68	Moderate to strong relationship
Content Marketing	0.62	Moderate relationship
SEM	0.53	Moderate relationship
Digital	0.39	Weak relationship
Radio	0.22	Weak relationship
Other	0.21	Weak relationship

Online Marketing and Affiliates showed the strongest correlation with revenue. Sponsorship, despite being the highest spend channel, had a lower correlation efficiency, suggesting it was approaching diminishing returns territory.

6. Model Results

6.1 Model Performance

Metric	Value
R-squared	0.977
Adjusted R-squared	0.950
Mean Absolute Error	1.56 Cr

Root Mean Squared Error	RMSE reported in Cr units
Mean Absolute Percentage Error	54.91% (inflated by Aug 2015 outlier)
Features Used	6
Observations	12 months

The model achieved strong predictive performance with an R-squared of 0.977. The high MAPE value is entirely driven by the August 2015 near-zero revenue month acting as an outlier in percentage error calculations. The model otherwise tracks actual revenue closely across all other months.

6.2 Channel Revenue Contributions

Feature	Revenue Contribution (%)
NPS Score	43.4
Affiliates	19.1
Sponsorship	17.7
Online Marketing	15.7
TV	3.2
Promo Days	0.9

NPS (Net Promoter Score) was the single largest contributor to modeled revenue at 43.4%, indicating that customer satisfaction plays a critical role in driving sales at DT MART. Among marketing channels, Affiliates and Sponsorship led at 19.1% and 17.7% respectively.

7. Diminishing Returns Analysis

Logarithmic saturation curves were fitted for the top four marketing channels. The following patterns were observed:

Channel	Saturation Point	Insight
Sponsorship	Around 400M spend	Revenue has plateaued. Further spend generates minimal incremental return
Online Marketing	Around 100M spend	Returns begin to flatten. Channel still has some headroom at current levels
Affiliates	Around 50M spend	Returns flatten beyond 50M. Channel is most efficient at current spend levels
TV	Early saturation	Flattest curve overall. Returns diminish quickly even at low spend

The analysis confirms that Sponsorship is the most saturated channel and that Affiliates and Online Marketing offer the steepest return curves at their current investment levels.

8. Budget Optimization Recommendations

Channel	Current Spend (M)	Current %	Optimized %	Optimized Spend (M)
Sponsorship	3654M (approx)	54.96%	9.43%	Reduce significantly
Online Marketing	1937M (approx)	29.13%	15.77%	Moderate reduction
Affiliates	614M (approx)	9.23%	60.72%	Major increase
TV	444M (approx)	6.68%	14.08%	Moderate increase

The key recommendation is to significantly reduce Sponsorship spend and reallocate the freed budget toward Affiliates. Affiliates currently receives only 9.23% of the top four channel budget but delivers 19.1% of revenue contribution, making it the most underinvested and highest efficiency channel in the mix.

A moderate increase in TV spend is also recommended as the channel has not yet reached full saturation at its current investment level.

9. Key Insights

- Affiliates is the most cost-efficient marketing channel. It delivers 19.1% of revenue contribution from only 9.23% of total budget, giving it the highest return on investment in the mix.
- Sponsorship is significantly over-funded. It consumes 54.96% of the top-four channel budget but contributes only 17.7% of modeled revenue. The saturation curve confirms the channel has reached diminishing returns.
- Online Marketing is reliable but also approaching saturation. With a 0.89 correlation to revenue and moderate contribution of 15.7%, it remains an important channel but should not receive further budget increases.
- NPS is the strongest single predictor of revenue, accounting for 43.4% of model contribution. Investments in customer experience and satisfaction have a measurable impact on sales performance.
- The August 2015 anomaly should be investigated. Near-zero revenue in that month suggests a data recording issue rather than a true business event and should be flagged for data quality review.

10. Tools and Techniques

Category	Details
Language	Python 3
Data Processing	Pandas, NumPy
Statistical Modeling	Statsmodels (OLS Regression)
Machine Learning	Scikit-learn (metrics, preprocessing)
Visualization	Matplotlib, Seaborn
Environment	Google Colab
Data Source	Kaggle (DT MART Market Mix Modeling dataset)
Techniques	OLS Regression, Log Transformation, Pearson Correlation, Diminishing Returns Modeling, Feature Engineering